



WhatsApp bans 23 lakh accounts

Meta-owned WhatsApp on Saturday said it banned over 23 lakh accounts in India in August.
<https://dgit.in/oct22-03>



Google receives 37k complaints

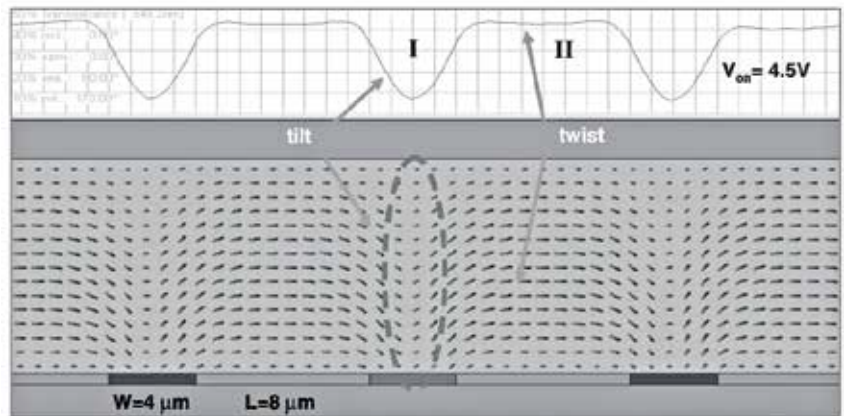
Google received 37,282 complaints from Indian users in the month of August.
<https://dgit.in/oct22-04>

and FFS displays are both very similar and were designed to overcome the problems of poor viewing angles from Twisted Nematic or TN LCD panels. In both IPS and FFS LCD displays, the initial alignment of the LC molecules is in the plane of the cell. When a bias voltage is applied, the generated electric field rotates the LC molecules in the plane, resulting in the transmittance of the backlight. However, the major difference between IPS LCD and FFS LCD displays is in the placement of electrodes. An FFS Cell has more electrodes packed closer to each other as compared to an IPS cell. This results in an extra horizontal component of an electric field or a stronger fringe field above the electrodes. An IPS cell requires a wide distance (l) between electrodes with minimal electrode width (w) to achieve high transmittance, but this also results in high operating voltage. At the edges of the electrode in an IPS cell, there is only one vertical component of the electric field, and thus the LC molecules just above the electrode do not rotate effectively. This results in minimum transmittance in the region right above the electrodes.

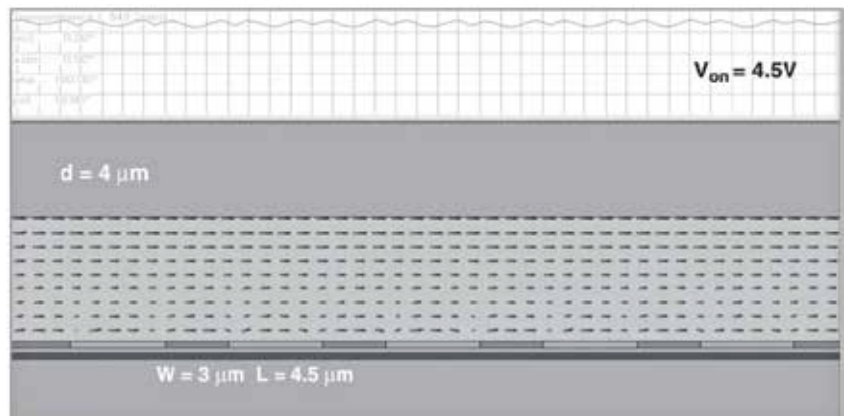
To increase transmittance in an IPS cell, manufacturers can apply higher bias voltage. Manufacturers can choose an LC material with a lower response time. In other words, there is a tradeoff involved between transmittance, response time, and power consumption.

By packing more electrodes closely, FFS displays counter these IPS LCD drawbacks while also retaining wide viewing angles. When a bias voltage is applied, the resulting fringe electric field has both horizontal and vertical components. Since the electrodes are closely spaced, the horizontal electric field (which is missing in the IPS cell) is present and it is strongest on the edges of the electrodes. The fringing fields are able to twist the LC molecules above the electrodes. This ultimately results in uniform transmittance across the entire cell.

Of course, we are oversimplifying things a bit to avoid making this article overly technical and mathematical, but

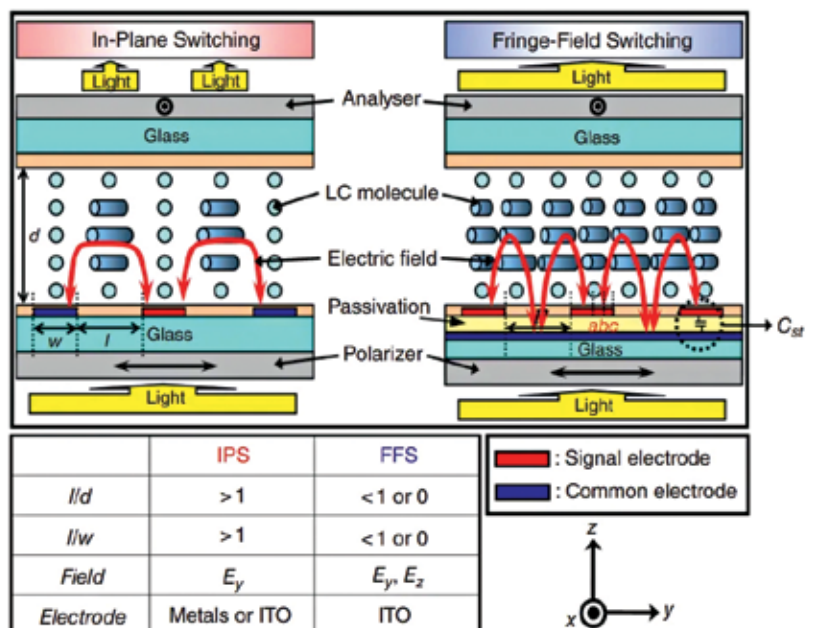


IPS Cell - transmittance is low in the region right above the electrode (Region I << Region II)



FFS Cell - uniform transmittance in the FFS Cell

FRINGE-FIELD SWITCHING (FFS) TECHNOLOGIES



In the FFS device, the pixel addressing electrodes and common electrodes are fabricated as different layers with a passivation layer between them